A FIXED-MASK PERMUTATION MODEL FOR THE MÖBIUS SUMMATORY PATH
EXACT FINITE MOMENTS, QUADRATIC PATH MASS, AND COMPUTATIONAL PLACEMENT

ABSTRACT. For a fixed cutoff N, the nonzero values of the Möbius function form a finite sign word on the squarefree support. We study the uniform orbit obtained by permuting those signs while preserving the support and the two sign counts. This is an exchangeable hypergeometric bridge with the same endpoint $M(N)$ as the observed word. For the normalized logarithmic quadratic path mass

$$\mathcal{Q}_{N_0,N}(w) = \frac{1}{\log(N/N_0)} \sum_{N_0<n\le N} \frac{R_w(n)^2}{n^2}, \qquad R_w(n) = \sum_{k\le n} w(k), \tag{0.1}$$

we derive exact finite-class formulae for the mean, the covariance kernel of squared partial sums, and the variance. We also give an exact bilinear decomposition, an opposite-sign transposition calculus, the reflection degeneracy of completed excursions, and the exact first moment of the number of returns to zero.

For the identified Möbius data file through $N = 455{,}052{,}511$, the reported full-cutoff values are

$$\mathcal{Q}_{1000,N}(\mu) = 0.029334, \qquad \mathbb{E}_\pi \mathcal{Q}_{1000,N}(\pi) = 0.56628, \qquad \rho_Q(N) = 0.0518. \tag{0.2}$$

The inequality between the first two quantities is a finite, data-dependent computer-assisted statement with a declared forward-error audit; it is not an asymptotic theorem. The approximate five-percent scale is largely explained, under standard additional hypotheses, by the classical spectral second moment of the Mertens function. Thus the class-relative finite comparison is the contribution of the model, whereas the small absolute quadratic scale is not claimed as a new phenomenon of analytic number theory. Reported excursion, conditional-choice, and blind spectral experiments are included with their observed outcomes and inferential limitations. No result in the paper proves the Riemann Hypothesis or the weak Mertens conjecture.

CONTENTS

1. Purpose, claims, and novelty boundary 2
1.1. What is established 3
1.2. What is new, and what is classical 3
1.3. Statistical status 3
2. The fixed-mask permutation class 3
2.1. Support, signs, and class hierarchy 3
2.2. Exact exchangeable moments 4
2.3. The critical square-root scale 5
3. Normalized logarithmic quadratic path mass 7
3.1. Definition and plateau representation 7
3.2. Relation to quadratic forms under permutation 7
3.3. Exact finite-class mean and variance 7
3.4. The exact order-dependent term 8
3.5. Exact change under a transposition 9
3.6. Occupation interpretation 9
3.7. Support-time coefficients and a feedback heuristic 10
4. Excursion degeneracy and exact return counts 11

Date: Version 3.

4.1. Reflection orbit 11
4.2. Expected number of returns 11
5. Finite placement of the Möbius word 12
5.1. Lower-tail sampling 12
5.2. Continuum diagnostics 13
6. Classical analytic meaning of the quadratic scale 14
6.1. Full-prefix and truncated normalizations 14
6.2. Weak Mertens and RH 14
6.3. Conditional spectral constant 16
7. Structural experiments beyond quadratic mass 17
7.1. Excursion orientation and shape 17
7.2. Scale partition and slow-envelope controls 17
7.3. Conditional-choice analysis 18
7.4. Blind recovery of logarithmic frequencies 18
8. A route toward genuine arithmetic selection 19
8.1. The exact selector already known 19
8.2. Why the finite class cannot classify RH behavior 19
8.3. The missing inequality 19
9. Conclusion 20
Appendix A. Numerical tables 20
A.1. Fixed-class quadratic comparison 20
A.2. Uniform-permutation ladder 21
A.3. Support-time coefficients 21
A.4. Excursion diagnostics 22
A.5. Partition and conditional-choice diagnostics 23
A.6. Blind frequency audit 23
Appendix B. Reproducibility specification 23
B.1. Required archive structure 23
B.2. Canonical source inventory 24
B.3. Input definition and integrity checks 25
B.4. Exact-layer configuration 25
B.5. Floating-point audit 26
B.6. Mandatory exact self-tests 26
B.7. Uniform sampling protocol 26
B.8. Excursion and conditional-choice protocols 27
B.9. Blind spectral protocol 27
B.10. Separate spectral-second-moment protocol 28
B.11. Raw output inventory and acceptance criteria 28
References 29

1. Purpose, claims, and novelty boundary

Let μ denote the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad Q(x) = \sum_{n \le x} \mu(n)^2. \tag{1.1}$$

The model asks a finite conditional question: after retaining the exact squarefree support and the exact numbers of positive and negative values, how does the order of the Möbius signs compare

with all other admissible orders? It does not replace multiplicative number theory by a random-sign hypothesis. It supplies a null class in which the information removed and the information retained are explicit.

1.1. **What is established.** The mathematical core consists of identities valid for every finite fixed-support sign orbit:

(i) the exact mean of (0.1);
(ii) a factorization of $\mathrm{Cov}(T_j^2, T_k^2)$ and an $O(m)$ formula for the exact variance;
(iii) an exact decomposition into a class-invariant diagonal term and an order-dependent bilinear term;
(iv) an exact formula for the change under one opposite-sign transposition;
(v) the reflection orbit of completed excursions and an exact formula for the mean number of zero returns;
(vi) the diffusive, Brownian-bridge critical scale of the null class.

The numerical component places the observed finite Möbius word in this class at fourteen cutoffs. The strongest data statement is the finite ratio in (0.2), not a limit as $N \to \infty$.

1.2. **What is new, and what is classical.** The finite-population bridge, its concentration inequalities, and its Brownian-bridge limit belong to standard probability theory [13, 22, 21, 1]. The weak Mertens conjecture, the conditional spectral second moment, and negative moments of ζ' are classical analytic number theory [24, 19, 9]. No priority is claimed for a general finite-population identity in isolation.

General quadratic forms under permutation and quadratic-assignment methods already possess an exact-moment literature [12, 14, 26]. The contribution asserted here is therefore narrower: the particular fixed-mask, fixed-count orbit for the Möbius word; the factorization and one-pass variance formula for its nested-max partial-sum kernel; the local ordering calculus; and the finite placement of the observed word against that exactly specified class. No claim is made that exact moments of a generic permuted quadratic form are new.

Remark 1.1 (The meaning of the five-percent observation). The finite value 0.0518 is not merely a check that a known conjecture holds on a finite interval. It compares two different objects: the observed Möbius path and the exact mean of its own fixed-mask, fixed-endpoint orbit. That particular comparison appears to be specific to the present model. However, the numerator's scale near 0.029 is conditionally explained by the classical spectral constant in Section 6. Consequently, the class-relative normalization is the distinctive part; the small absolute second moment is not presented as an independent discovery.

Warning 1.2 (No uniqueness claim). A single quadratic functional cannot select the Möbius word uniquely. It is constant on an exponentially large excursion-reflection orbit and many other words have smaller values. Section 8 gives the exact arithmetic functional that does select μ, explains why that selection is tautological from Möbius inversion, and states what additional inequality would be needed to connect selection to RH-scale growth.

1.3. **Statistical status.** The permutation ladder, excursion samples, conditional-choice analyses, and blind spectral fit are reported computations. Exact zero-hit bounds use the one-sided Clopper–Pearson construction [3]. The conditional-choice p-values and the blind frequency matches remain exploratory until their raw outputs and complete selection pipelines are rerun independently. Appendix B specifies the required reproducibility package without executing it.

2. The fixed-mask permutation class

2.1. **Support, signs, and class hierarchy.** Fix $N \geq 1$ and write

$$A_N = \{n \leq N : \mu(n) \neq 0\} = \{a_1 < a_2 < \cdots < a_m\}, \qquad m = Q(N). \tag{2.1}$$

Let p and q be the numbers of $+1$ and -1 values on A_N. Then

$$p+q=m, \qquad p-q=S=M(N), \qquad p=\frac{m+S}{2}, \quad q=\frac{m-S}{2}. \tag{2.2}$$

Definition 2.1 (Fixed-mask and fixed-count classes)**.** Define

$$\Omega_N^{(0)}=\{w\in\{-1,0,1\}^N : \operatorname{supp}(w)=A_N\}, \tag{2.3}$$

$$\Omega_N^{(1)}=\left\{w\in\Omega_N^{(0)} : \#\{n : w(n)=+1\}=p\right\}. \tag{2.4}$$

$$\Omega_N^{(2)}=\left\{w\in\Omega_N^{(1)} : \left|\sum_{k\le n} w(k)\right|=|M(n)| \text{ for every } n\le N\right\}. \tag{2.5}$$

Equivalently, $\Omega_N^{(1)}$ is the orbit of $(\mu(1),\dots,\mu(N))$ under the symmetric group acting on the values at the ordered support positions.

The two defining constraints are the zero mask and the sign multiset. The endpoint follows from them; it is not a third constraint. The first two cardinalities are

$$|\Omega_N^{(0)}|=2^m, \qquad |\Omega_N^{(1)}|=\binom{m}{p}. \tag{2.6}$$

The entire function $Q(n)$, $n\le N$, and the quantities p,q,S are class invariants. The intermediate summatory path is not.

For $w\in\Omega_N^{(1)}$, let

$$\xi_j=w(a_j), \qquad T_j=\sum_{i=1}^{j}\xi_i, \qquad T_0=0. \tag{2.7}$$

Then

$$R_w(n)=T_{Q(n)}, \qquad R_w(N)=T_m=S. \tag{2.8}$$

If

$$K_N=\#\{1\le j\le m : T_j(\mu)=0\}, \tag{2.9}$$

then Proposition 4.1 proves

$$\Omega_N^{(0)}\supseteq\Omega_N^{(1)}\supseteq\Omega_N^{(2)}, \qquad 2^m\ge\binom{m}{p}\ge 2^{K_N}, \qquad |\Omega_N^{(2)}|=2^{K_N}. \tag{2.10}$$

Both inequalities are strict for the reported data at $N=5\cdot 10^6$, where the base-ten logarithms of the middle and final cardinalities are $915017.333\ldots$ and $1810.996\ldots$, respectively. At each level, the uniformization discards the divisor relations except for the invariants explicitly frozen by that level. The uniform measure on $\Omega_N^{(1)}$ is sampling without replacement from a population containing p values $+1$ and q values -1. Hence $(T_j)_{0\le j\le m}$ is an exchangeable hypergeometric bridge; the finite-population moment framework is standard [4].

2.2. **Exact exchangeable moments.** Assume first that $m\ge 4$. For distinct indices define

$$\mu_1=\mathbb{E}\xi_a=\frac{S}{m}, \tag{2.11}$$

$$\mu_2=\mathbb{E}(\xi_a\xi_b)=\frac{S^2-m}{m(m-1)}, \tag{2.12}$$

$$\mu_4=\mathbb{E}(\xi_a\xi_b\xi_c\xi_d)=\frac{S^4-6mS^2+8S^2+3m^2-6m}{m(m-1)(m-2)(m-3)}. \tag{2.13}$$

The identities follow by expanding powers of $\sum_{j=1}^{m} \xi_j = S$, using $\xi_j^2 = 1$, and exploiting exchangeability. In particular,

$$\mathbb{E}T_j^2 = j + j(j-1)\mu_2. \tag{2.14}$$

For $m < 2$, the second mixed moment is unnecessary and the class is enumerated directly. For $2 \le m < 4$, formula (2.12) remains available but fourth-order quantities are evaluated by direct finite enumeration; no undefined quotient is used.

Theorem 2.2 (Factored covariance of squared partial sums)**.** *For $m \ge 4$ and $1 \le j \le k \le m$,*

$$\operatorname{Cov}(T_j^2, T_k^2) = \phi(j)\psi(k), \qquad \phi(j) = j(j-1), \tag{2.15}$$

where

$$\psi(k) = 2 + 4\mu_2(k-2) + \mu_4(k-2)(k-3) - \mu_2^2 k(k-1). \tag{2.16}$$

In particular, $\psi(m) = 0$, as required by the deterministic endpoint $T_m = S$.

Proof. Write $T_k = T_j + U_{j,k}$, expand $T_j^2 T_k^2$, and group monomials by the number of distinct indices. Exchangeability replaces all products with two distinct indices by μ_2 and all products with four distinct indices by μ_4. Subtracting $\mathbb{E}T_j^2 \mathbb{E}T_k^2$ and collecting the coefficients gives (2.15)–(2.16). The complete enumeration audit specified in Appendix B checks all pairs for every sign class with $m \le 8$. □

2.3. **The critical square-root scale.** Put

$$U_j = T_j - \frac{jS}{m}, \qquad \sigma_m^2 = 1 - \frac{S^2}{m^2}. \tag{2.17}$$

For each fixed j, Hoeffding's comparison theorem for sampling without replacement gives

$$\mathbb{P}\left(|U_j| \ge \lambda\right) \le 2\exp\left(-\frac{\lambda^2}{2j}\right). \tag{2.18}$$

Applying a union bound, rather than claiming a sharp maximal inequality, yields

$$\mathbb{P}\left(\max_{1\le j\le m} |U_j| > \lambda\right) \le 2m\exp\left(-\frac{\lambda^2}{2m}\right). \tag{2.19}$$

Thus, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left(\max_{j\le m} |U_j| > m^{1/2+\varepsilon}\right) \le 2m\exp\left(-\frac{m^{2\varepsilon}}{2}\right). \tag{2.20}$$

Lemma 2.3 (Uniform square integrability of the bridge maximum)**.** *For every nondegenerate fixed-count sign class with $m \ge 3$,*

$$\mathbb{E}_\pi\left[\left\{\frac{\max_{0\le j\le m}|U_j|}{\sigma_m\sqrt{m}}\right\}^2\right] \le 32. \tag{2.21}$$

Consequently the normalized maxima are uniformly integrable along every sequence of such classes.

Proof. Let $X_i = \xi_i - S/m$, let $\mathcal{F}_j$ be the first j revealed signs, and put $h = \lfloor m/2 \rfloor$. Since the unrevealed centered population has sum $-U_j$,

$$\mathbb{E}(X_{j+1} \mid \mathcal{F}_j) = -\frac{U_j}{m-j}. \tag{2.22}$$

It follows that

$$Y_j = \frac{U_j}{m-j}, \qquad 0 \le j < m, \tag{2.23}$$

is a martingale. Doob's L^2 inequality and

$$\mathbb{E}U_h^2 = \sigma_m^2 \frac{h(m-h)}{m-1} \tag{2.24}$$

give

$$\mathbb{E}\max_{0\le j\le h} U_j^2 \le m^2\mathbb{E}\max_{0\le j\le h} Y_j^2 \le 4m^2\mathbb{E}Y_h^2 \le 8\sigma_m^2 m. \tag{2.25}$$

Apply the same argument to the reversed centered sequence. Since its partial sums satisfy $\widetilde{U}_r = -U_{m-r}$, the second half is bounded by

$$\mathbb{E}\max_{h\le j\le m} U_j^2 \le 16\sigma_m^2 m. \tag{2.26}$$

The square of the full maximum is at most the sum of the two half-maxima. This gives the stated, slightly relaxed constant 32; the finitely many small boundary cases are checked directly. □

Proposition 2.4 (Finite-population bridge limit)**.** *Suppose $m \to \infty$ and $|S|/m \to s_0 < 1$. The polygonal interpolation of*

$$\mathbb{T}_m(t) = \frac{T_{\lfloor mt\rfloor} - (\lfloor mt\rfloor/m)S}{\sigma_m\sqrt{m}}, \qquad 0 \le t \le 1, \tag{2.27}$$

converges weakly to a standard Brownian bridge B°. Consequently,

$$\frac{1}{\sigma_m\sqrt{m}}\max_{j\le m}|U_j| \Longrightarrow \sup_{0\le t\le 1}|B^\circ(t)|. \tag{2.28}$$

$$\mathbb{E}\max_{j\le m}|U_j| \sim \sigma_m\sqrt{\frac{\pi}{2}}\log 2\,\sqrt{m}. \tag{2.29}$$

Proof. The weak convergence is the finite-population functional central limit theorem; see [21]. The continuous-mapping theorem gives (2.28). Lemma 2.3 supplies uniform integrability, so convergence of the first moments follows. The limit distribution and its first moment use

$$\mathbb{P}\left(\sup_t|B^\circ(t)| \le x\right) = \sum_{r\in\mathbb{Z}}(-1)^r e^{-2r^2x^2}, \tag{2.30}$$

$$\mathbb{E}\sup_t|B^\circ(t)| = \sqrt{\frac{\pi}{2}}\log 2 = 0.868731\ldots, \tag{2.31}$$

as recorded, for example, in [7]. □

At $j = \lfloor m/2\rfloor$, the hypergeometric local central limit theorem gives, for every fixed $c > 0$,

$$\mathbb{P}\left(|U_{\lfloor m/2\rfloor}| > c\sigma_m\sqrt{m}\right) \longrightarrow 2\overline{\Phi}(2c) > 0. \tag{2.32}$$

Equations (2.20) and (2.32) show that $m^{1/2}$ is the critical finite scale of the centered class. This is the same exponent that occurs in the classical RH criterion, but the logical statements are different.

Warning 2.5 (Endpoint and coupling)*.* The deterministic endpoint satisfies

$$\left|\max_j|T_j| - \max_j|U_j|\right| \le |S|, \qquad \max_j|T_j| \ge |S|. \tag{2.33}$$

A concentration statement for U_j does not control $S = M(N)$. Moreover, the finite classes at different values of N are not a single canonically coupled random sequence. The law of the iterated logarithm is therefore not a prediction for one draw from $\Omega_N^{(1)}$; it concerns an almost-sure envelope along a coupled infinite sequence [10].

Warning 2.6 (Why typicality does not imply RH)*.* At $N = 5\cdot 10^6$, a designated member has uniform mass

$$|\Omega_N^{(1)}|^{-1} = 10^{-915017.333\ldots}. \tag{2.34}$$

An event containing a $1 - o(1)$ fraction of a finite class need not contain that designated member. In addition, the class deliberately discards the divisor relations that define μ. No concentration inequality over the class transfers by itself to the Möbius word.

3. Normalized logarithmic quadratic path mass

3.1. **Definition and plateau representation.** Fix $1 \le N_0 < N$ and put

$$L_N = \log(N/N_0). \tag{3.1}$$

Definition 3.1 (Normalized logarithmic quadratic path mass)**.** For a finite word w and its partial sums $R_w(n)$, define

$$\mathcal{Q}_{N_0,N}(w) = \frac{1}{L_N} \sum_{N_0<n\le N} \frac{R_w(n)^2}{n^2}. \tag{3.2}$$

This is a positive logarithmically weighted second moment of the summatory path. No physical interpretation is assumed.

For the fixed mask, define the plateau weights

$$w_j = w_j^{(N_0,N)} = \sum_{\substack{N_0<n\le N \\ Q(n)=j}} \frac{1}{n^2}, \qquad 1 \le j \le m. \tag{3.3}$$

Then the path functional is the quadratic form

$$L_N \mathcal{Q}_{N_0,N}(w) = \sum_{j=1}^{m} w_j T_j^2. \tag{3.4}$$

The weights depend only on the squarefree mask; all order dependence is in the bridge coordinates T_j.

3.2. **Relation to quadratic forms under permutation.** Let $\xi = (\xi_1, \dots, \xi_m)^{\mathsf{T}}$. Expanding the partial sums in (3.4) gives

$$L_N \mathcal{Q}_{N_0,N}(w) = \xi^{\mathsf{T}} \mathbf{K}_N \xi, \tag{3.5}$$

$$(\mathbf{K}_N)_{rs} = \sum_{j=\max(r,s)}^{m} w_j, \qquad 1 \le r, s \le m. \tag{3.6}$$

Thus the model is a quadratic form evaluated on a uniform permutation of a fixed multiset of signs. Permutation quadratic forms, quadratic-assignment statistics, and their exact moments have an established literature [12, 14, 26]. In particular, exact permutation moments of general quadratic forms are not claimed here as a new category of result.

The specific feature used below is the nested-max kernel (3.6). For this kernel and the fixed-count sign population, the covariance of squared prefix sums factorizes on the upper triangle, reducing the variance to one prefix pass; the same prefix geometry also yields the local transposition formula in Section 3.5. These special simplifications, rather than the generic quadratic-form representation, delimit the finite probabilistic contribution.

3.3. **Exact finite-class mean and variance.** Define two mask sums

$$\Sigma_1(N_0, N) = \sum_{N_0<n\le N} \frac{Q(n)}{n^2} = \sum_{j=1}^{m} j w_j, \tag{3.7}$$

$$\Sigma_2(N_0, N) = \sum_{N_0<n\le N} \frac{Q(n)^2}{n^2} = \sum_{j=1}^{m} j^2 w_j. \tag{3.8}$$

For $m \ge 2$, let

$$A = \frac{m^2 - S^2}{m(m-1)} = 1 - \mu_2, \qquad B = \frac{S^2 - m}{m(m-1)} = \mu_2. \tag{3.9}$$

Theorem 3.2 (Exact mean over the complete class)**.** *For the uniform measure on* $\Omega_N^{(1)}$*,*

$$\overline{\mathcal{Q}}_N := \mathbb{E}_\pi \mathcal{Q}_{N_0,N}(\pi) = \frac{A\Sigma_1 + B\Sigma_2}{L_N} \tag{3.10}$$

$$= \frac{1}{L_N} \sum_{j=1}^{m} w_j\{j + \mu_2 j(j-1)\}. \tag{3.11}$$

Thus all cutoffs can be evaluated in one streaming pass, because Σ_1 *and* Σ_2 *are cumulative while* A *and* B *depend only on the terminal values* m *and* S*.*

Proof. Substitute (2.14) into (3.4). The identity $\mathbb{E}T_j^2 = Aj + Bj^2$ then gives (3.10); expanding $A = 1 - \mu_2$ gives (3.11). □

Theorem 3.3 (Exact variance in one prefix pass)**.** *Assume* $m \geq 4$*, and define*

$$P_k = \sum_{j<k} w_j \phi(j). \tag{3.12}$$

Then

$$\mathrm{Var}_\pi(\mathcal{Q}_{N_0,N}(\pi)) = \frac{1}{L_N^2} \sum_{k=1}^{m} w_k \psi(k)\{w_k\phi(k) + 2P_k\}. \tag{3.13}$$

The formula uses $O(m)$ *operations and* $O(1)$ *running state once the weights are streamed.*

Proof. Expand the variance of $\sum_j w_j T_j^2$, use symmetry of covariance, and apply Theorem 2.2:

$$L_N^2 \,\mathrm{Var}(\mathcal{Q}) = \sum_{k=1}^{m} w_k^2 \phi(k)\psi(k) + 2 \sum_{1\leq j<k\leq m} w_j w_k \phi(j)\psi(k) \tag{3.14}$$

$$= \sum_{k=1}^{m} w_k \psi(k) \left\{ w_k\phi(k) + 2\sum_{j<k} w_j \phi(j) \right\}. \tag{3.15}$$

This is (3.13). □

Remark 3.4 (Exact formula versus numerical evaluation). Equations (3.10) and (3.13) are exact finite identities. Decimal values reported later are binary64 evaluations with explicit forward-error bounds; they are not directed-rounding interval evaluations.

3.4. **The exact order-dependent term.** The elementary square increment is

$$T_j^2 - T_{j-1}^2 = 1 + 2\xi_j T_{j-1}. \tag{3.16}$$

Define the future plateau weight

$$W_b = \sum_{j=b}^{m} w_j. \tag{3.17}$$

Proposition 3.5 (Diagonal term plus signed restoration)**.** *Every member of* $\Omega_N^{(1)}$ *satisfies*

$$L_N \mathcal{Q}_{N_0,N}(w) = \Sigma_1(N_0, N) + 2\mathcal{J}_N(w), \tag{3.18}$$

$$\mathcal{J}_N(w) = \sum_{b=2}^{m} W_b \xi_b T_{b-1}. \tag{3.19}$$

The first term is a mask invariant. Consequently the complete class is ordered by the single scalar $\mathcal{J}_N$*.*

Proof. Insert (3.16) into $T_j^2 = \sum_{b\leq j}(T_b^2 - T_{b-1}^2)$, multiply by w_j, and interchange the two finite sums. The diagonal contribution is $\sum_j j w_j = \Sigma_1$. □

Exchangeability also gives

$$\mathbb{E}_\pi(\xi_b T_{b-1}) = (b-1)\mu_2, \tag{3.20}$$

and hence

$$\mathbb{E}_\pi \mathcal{J}_N = \mu_2 \sum_{b=2}^{m} (b-1) W_b. \tag{3.21}$$

For the nearly balanced classes in the data, μ_2 is close to $-1/(m-1)$. The much smaller observed value of $\mathcal{Q}$ therefore means that the ordered products $\xi_b T_{b-1}$, with their future weights, are substantially more negative than their exchangeable-class average.

3.5. **Exact change under a transposition.** Let $a < b$ be support indices carrying opposite signs. Swapping them preserves the mask, sign counts, and endpoint.

Proposition 3.6 (Opposite-sign transposition formula)**.** *If $\xi_a = +1$ and $\xi_b = -1$, then*

$$L_N\{\mathcal{Q}(\pi \circ \tau_{a,b}) - \mathcal{Q}(\pi)\} = 4 \sum_{j=a}^{b-1} w_j (1 - T_j). \tag{3.22}$$

If $\xi_a = -1$ and $\xi_b = +1$, then

$$L_N\{\mathcal{Q}(\pi \circ \tau_{a,b}) - \mathcal{Q}(\pi)\} = 4 \sum_{j=a}^{b-1} w_j (1 + T_j). \tag{3.23}$$

Proof. In the first case the path decreases by 2 on $a \le j < b$ and is unchanged elsewhere, so $(T_j - 2)^2 - T_j^2 = 4(1 - T_j)$. The second case is analogous. □

With

$$\overline{T}_w[a,b) = \frac{\sum_{j=a}^{b-1} w_j T_j}{\sum_{j=a}^{b-1} w_j}, \tag{3.24}$$

a $+,-$ transposition decreases the quadratic mass exactly when $\overline{T}_w[a,b) > 1$, and a $-,+$ transposition decreases it exactly when $\overline{T}_w[a,b) < -1$. Thus a local minimum under all admissible transpositions is characterized by these two orientation-sensitive families of inequalities. They are not equivalent to a single condition $|\overline{T}_w[a,b)| \le 1$ over every interval, because not every interval has both endpoint orientations.

The lower set determined by the observed word is therefore

$$\mathcal{A}_N = \{\pi \in \Omega_N^{(1)} : \mathcal{Q}_{N_0,N}(\pi) \le \mathcal{Q}_{N_0,N}(\mu)\} = \{\pi : \mathcal{J}_N(\pi) \le \mathcal{J}_N(\mu)\}. \tag{3.25}$$

Finding the best single transposition takes $O(m)$ time using running extrema of weighted prefixes. Counting every improving pair is a different problem and generally requires more than that single pass.

3.6. **Occupation interpretation.** Put $d = 1/\zeta(2) = 6/\pi^2$, the standard squarefree density [18]. Since the j-th support plateau is centered near $n = j/d$ and has average integer length $1/d$, one expects

$$w_j \sim \frac{d}{j^2}. \tag{3.26}$$

This gives the useful approximation

$$\mathcal{Q}_{N_0,N}(w) \approx d \left\langle \frac{T_j^2}{j} \right\rangle_{\log j}. \tag{3.27}$$

The exact probability measure behind this statement is

$$\nu_N(\{j\}) = \frac{j w_j}{D_N}, \qquad D_N = \sum_{j=1}^{m} j w_j = \Sigma_1(N_0, N). \tag{3.28}$$

Its normalized quadratic occupation level is

$$A_N^{\mathrm{occ}} = \int \frac{T_j^2}{j}\, d\nu_N(j) = \frac{L_N \mathcal{Q}_{N_0,N}(w)}{\Sigma_1(N_0,N)}. \tag{3.29}$$

For every $\lambda > 0$, Markov's inequality gives the exact consequence

$$\nu_N\left\{ j : \frac{T_j^2}{j} > \lambda \right\} \le \frac{A_N^{\mathrm{occ}}}{\lambda}. \tag{3.30}$$

At the full cutoff, the density-normalized approximation $a_N = \zeta(2)\mathcal{Q}(\mu) = 0.048252$ implies, at the descriptive level, that the path lies below $\sqrt{j}$ for at least 95.2 percent of logarithmic support time and below $2\sqrt{j}$ for at least 98.8 percent. The exact percentages use A_N^{occ} rather than a_N; the difference is the finite mask correction in (3.28).

3.7. **Support-time coefficients and a feedback heuristic.** For an interval $u < j \le v$, define

$$c_{u,v} = \frac{1}{v-u} \sum_{j=u+1}^{v} \frac{T_j^2}{j}. \tag{3.31}$$

Under an explicitly specified ensemble, one may also define

$$c(K) = \frac{\mathrm{Var}(T_K)}{K}. \tag{3.32}$$

These are distinct quantities. The algebraic normalization

$$a_N = \zeta(2)\mathcal{Q}_{N_0,N}(\mu) \tag{3.33}$$

is not an independent measurement of either coefficient.

Heuristic 3.7 (Critical restoring drift)**.** This is a heuristic argument. Suppose a process obeys

$$\mathbb{E}(\xi_{j+1} \mid T_j) \approx -\frac{2a}{j} T_j. \tag{3.34}$$

Writing $V_j = \mathbb{E}T_j^2$ gives

$$V_{j+1} - V_j \approx 1 - \frac{4a}{j} V_j. \tag{3.35}$$

A solution $V_j \sim cj$ then satisfies

$$c = \frac{1}{1+4a}, \qquad a = \frac{c^{-1}-1}{4}. \tag{3.36}$$

The power j^{-1} is critical: a drift $j^{-\gamma}$ formally gives $V_j/j \to 1$ for $\gamma > 1$, $V_j/j \to 0$ for $\gamma < 1$, and permits a strict constant in $(0,1)$ only for $\gamma = 1$. This model can explain how a cumulative suppression coexists with a null nearest-neighbour persistence statistic, but (3.36) is not an identity for the Möbius word.

The scale measurements in Table 3 fluctuate between approximately 0.036 and 0.068, without monotone convergence. A sinusoidal fit in $\log_2 K$ is reported in Appendix A; its out-of-sample performance does not establish a predictive law.

4. Excursion degeneracy and exact return counts

4.1. **Reflection orbit.** Let

$$0 = \tau_0 < \tau_1 < \cdots < \tau_K \tag{4.1}$$

be the successive completed zero-to-zero excursions of the support-time path. Write the r-th completed excursion as

$$T|_{[\tau_{r-1},\tau_r]} = \varepsilon_r H_r, \qquad \varepsilon_r \in \{-1,+1\}, \qquad H_r \geq 0. \tag{4.2}$$

Proposition 4.1 (Pointwise absolute-path orbit)**.** *Reflecting any subset of the K completed excursions preserves the zero mask, the two sign counts, the endpoint, and every value $R_w(n)^2$. Moreover, the resulting orbit exhausts the pointwise absolute-path class:*

$$\mathcal{O}_N(\mu) = \Omega_N^{(2)}, \qquad |\mathcal{O}_N(\mu)| = |\Omega_N^{(2)}| = 2^K, \qquad \mathcal{Q}_{N_0,N}(w) = \mathcal{Q}_{N_0,N}(\mu) \quad (w \in \mathcal{O}_N(\mu)). \tag{4.3}$$

If $M(N) \neq 0$, the terminal incomplete excursion after the last return is not reflected, because its reflection would change the fixed endpoint.

Proof. A completed excursion contains equally many positive and negative steps. Reflection changes every sign in that segment, preserves those counts, and replaces the path by its negative only within the segment. Independent choices therefore give 2^K distinct words in $\Omega_N^{(2)}$.

Conversely, take $w \in \Omega_N^{(2)}$. Between two consecutive zeros τ_{r-1} and τ_r, the reference path has no interior zero. Because both paths have unit increments in support time and the same absolute value at every support index, the sign relating them cannot change inside that interval. Hence the restriction of w to the excursion is either the original sign block or its complete reflection. These alternatives are independent across completed excursions. On a terminal incomplete segment, the fixed endpoint $T_m = S \neq 0$ forces the original orientation. Thus there are no further elements of $\Omega_N^{(2)}$, proving exhaustivity and the asserted equality. When $S = 0$, the terminal segment is itself the last completed excursion. □

Remark 4.2 (Consequence for selection)*.* The reflection orbit describes degeneracy after the absolute path has already been fixed. It does not explain why the observed absolute path has small quadratic mass. It proves instead that no functional depending only on $R_w(n)^2$ can select a unique orientation.

4.2. **Expected number of returns.** For $w \in \Omega_N^{(1)}$, define

$$Z_N(w) = \#\{1 \leq j \leq m : T_j(w) = 0\}. \tag{4.4}$$

Proposition 4.3 (Exact first moment of the return count)**.** *Under the uniform fixed-count measure,*

$$\mathbb{E}_\pi Z_N = \sum_{k=1}^{\min(p,q)} \frac{\binom{p}{k}\binom{q}{k}}{\binom{m}{2k}}. \tag{4.5}$$

Proof. A return is possible only at time $2k$. Among the $\binom{m}{2k}$ equally likely subsets of sampled signs, precisely $\binom{p}{k}\binom{q}{k}$ contain k signs of each type. Sum the indicators of the return events. □

At $N = 5 \cdot 10^6$, the exact and observed values reported in Appendix A are

$$\mathbb{E}_\pi Z_N = 1623.0496, \qquad Z_N(\mu) = 6016, \qquad \frac{Z_N(\mu)}{\mathbb{E}_\pi Z_N} = 3.7066. \tag{4.6}$$

The same path has longest completed-excursion fraction 0.11737, compared with a reported bridge-sample mean 0.5301. These are related descriptions of a path spending unusually little logarithmic mass far from zero; they are not independent events whose probabilities may be multiplied.

5. Finite placement of the Möbius word

Define the class-relative ratio and standardized displacement

$$\rho_Q(N) = \frac{\mathcal{Q}_{N_0,N}(\mu)}{\overline{\mathcal{Q}}_N}, \qquad Z_Q(N) = \frac{\mathcal{Q}_{N_0,N}(\mu) - \overline{\mathcal{Q}}_N}{\sqrt{\mathrm{Var}_\pi(\mathcal{Q}_{N_0,N})}}. \tag{5.1}$$

Proposition 5.1 (Finite separation)**.** *Let the input be the signed-byte file with 455,052,511 entries and SHA-256 digest*

$$\texttt{0e551f47d924ef6f3e348e131fdf53e0c677ad5e5471ef9d6249b3fd662e2ea4}. \tag{5.2}$$

Under the binary64 and execution assumptions specified in Appendix B, the fourteen cutoffs in Table 1 satisfy

$$\mathcal{Q}_{1000,N}(\mu) < \overline{\mathcal{Q}}_N. \tag{5.3}$$

At the complete cutoff,

$$\mathcal{Q}_{1000,455052511}(\mu) = 0.029334, \quad \overline{\mathcal{Q}}_{455052511} = 0.56628, \quad \rho_Q(455052511) = 0.0518. \tag{5.4}$$

At $N = 5 \cdot 10^6$, the higher-precision reported values are

$$\mathcal{Q}_{1000,N}(\mu) = 0.0295150019629, \quad \overline{\mathcal{Q}}_N = 0.548481916755, \quad \rho_Q(N) = 0.0538121696655. \tag{5.5}$$

Proof. The class mean is evaluated from Theorem 3.2; the observed value is a positive weighted sum. The streaming implementation uses blocked summation, compensation between blocks, and an a-priori forward-error propagation under the assumptions in Appendix B. For each cutoff the upper endpoint of the propagated interval for $\mathcal{Q}(\mu) - \overline{\mathcal{Q}}_N$ is negative. The smallest ratio of the absolute separation to the propagated bound is $5.75 \cdot 10^{10}$. Division gives (5.4) and (5.5). The proposition is conditional on the identified file and execution model; the exact symbolic identities do not depend on either. □

Remark 5.2 (What has been proved). Proposition 5.1 proves comparison with the exact class mean at fourteen finite cutoffs. It does not prove that $\rho_Q(N)$ is always near 0.05, that the ratio is monotone, or that it has a limit. Indeed the first tabulated value is 0.082224, and the later values exhibit small nonmonotone fluctuations.

At the complete cutoff,

$$m = 276{,}638{,}778, \qquad S = -5414, \qquad \max_{n \le N} |M(n)| = 6695. \tag{5.6}$$

Since $\sqrt{m} = 16632$ to the displayed precision, the observed maximum is

$$\frac{6695}{\sqrt{m}} = 0.402, \qquad \frac{6695}{0.868731\sqrt{m}} = 0.463. \tag{5.7}$$

The second denominator is the limiting mean maximum of a nearly balanced finite bridge. This is a finite comparison; it is not a law-of-the-iterated- logarithm statement.

5.1. **Lower-tail sampling.**

The exact lower-tail fraction is

$$P_Q(N) = \frac{\#\{\pi \in \Omega_N^{(1)} : \mathcal{Q}_{N_0,N}(\pi) \le \mathcal{Q}_{N_0,N}(\mu)\}}{|\Omega_N^{(1)}|}. \tag{5.8}$$

No closed form for (5.8) is established here. Uniform fixed-count sampling reported 412 hits in 300,000 draws at $N = 10^4$, 15 hits in 150,000 draws at $N = 3 \cdot 10^4$, and zero hits at each of the four larger sampled cutoffs. Full results and exact zero-hit bounds appear in Table 2.

Experiment 5.3 (Failure of a two-moment tail approximation)**.** A scaled chi-square law fitted to the exact mean and variance predicts, across the six cutoffs, expected hit counts

$$(20772,\ 6113.7,\ 2218.2,\ 477.88,\ 61.85,\ 23.242), \tag{5.9}$$

whereas the observed counts are

$$(412,\ 15,\ 0,\ 0,\ 0,\ 0). \tag{5.10}$$

At $N = 5 \cdot 10^6$, the approximation gives

$$P_Q^{(\chi^2)} = 0.0023242189, \qquad \mathbb{P}_{\chi^2}(H = 0) = 7.8397 \cdot 10^{-11}. \tag{5.11}$$

The binomial probability of an outcome at least this extreme is below $7.84 \cdot 10^{-11}$ in every row. Thus exact mean and variance do not determine the lower tail of this quadratic form.

Warning 5.4 (Mean separation is not a percentile). The ratio ρ_Q compares the observed value with the exact mean; it is not P_Q. Likewise, zero hits in R draws yield the exact one-sided 95-percent bound $P_Q < 1 - 0.05^{1/R}$, not an estimate of a probability of order 10^{-9}. At $N = 5 \cdot 10^6$, the class size is approximately $10^{915017.333}$, so exhaustive enumeration is impossible.

5.2. **Continuum diagnostics.** Put

$$d = \frac{1}{\zeta(2)}, \qquad y_N^2 = \frac{S^2}{m}, \qquad x_0 = e^{-L_N}. \tag{5.12}$$

Heuristic 5.5 (Finite-span Gaussian bridge)**.** This is a heuristic continuum replacement. Replacing the squarefree mask by density d and the hypergeometric bridge by a Gaussian bridge gives

$$\overline{\mathcal{Q}}_N^{\mathrm{cont}} = d\left[1 - (1 - y_N^2)\frac{1 - e^{-L_N}}{L_N}\right], \tag{5.13}$$

$$(\sigma_N^{\mathrm{cont}})^2 = \frac{d^2}{L_N^2}\Big[4L_N - 10 + 8(L_N + 1)e^{-L_N} + 2e^{-2L_N} + 4y_N^2\{1 - e^{-2L_N} - 2L_N e^{-L_N}\}\Big]. \tag{5.14}$$

Dropping only the exponentially small term in the mean gives

$$\overline{\mathcal{Q}}_N^{\mathrm{principal}} = d\left[1 - \frac{1 - y_N^2}{L_N}\right]. \tag{5.15}$$

When $y_N^2 = o(1)$,

$$\overline{\mathcal{Q}}_N^{\mathrm{cont}} = d\left(1 - \frac{1}{L_N}\right) + o(L_N^{-1}), \qquad (\sigma_N^{\mathrm{cont}})^2 = \frac{4d^2(L_N - 5/2)}{L_N^2} + o(L_N^{-2}). \tag{5.16}$$

At the complete cutoff, (5.15) gives 0.5662087354, while (5.13) gives 0.5662088271. The exact finite-class mean is 0.56628. The exact and continuum standard deviations are 0.30436 and 0.30433, respectively. These close numerical agreements validate the approximation at the measured scale but do not prove its asymptotic error term.

Heuristic 5.6 (Ornstein–Uhlenbeck lower-tail rate)**.** The Lamperti transform

$$V(t) = e^{t/2}W(e^{-t}), \qquad \mathrm{Cov}(V(s), V(t)) = e^{-|t-s|/2} \tag{5.17}$$

is a stationary Ornstein–Uhlenbeck process with unit variance. The Feynman–Kac calculation within the Donsker–Varadhan framework [6] gives

$$\Lambda(\lambda) = \frac{1}{2}\left(\frac{1}{2} - \sqrt{\frac{1}{4} - 2\lambda}\right), \qquad \lambda < \frac{1}{8}, \tag{5.18}$$

$$I(a) = \sup_{\lambda<1/8} \{\lambda a - \Lambda(\lambda)\} = \frac{(a-1)^2}{8a}. \tag{5.19}$$

Transferring this rate to the finite fixed-mask bridges requires an unproved large-deviation invariance principle and a boundary correction. The resulting prediction is

$$\log P_Q(N) \approx -L_N I(a_N) + \log \kappa(a_N, L_N), \qquad a_N = \zeta(2)\mathcal{Q}_{N_0,N}(\mu), \tag{5.20}$$

where the prefactor κ is not determined.

At $N = 5 \cdot 10^6$, the leading rate gives

$$a_N = 0.0485502, \qquad I(a_N) = 2.3307, \qquad \exp\{-L_N I(a_N)\} = 2.39 \cdot 10^{-9}. \tag{5.21}$$

At the complete cutoff the corresponding values are

$$a_N = 0.048252, \qquad I(a_N) = 2.3466, \qquad \exp\{-L_N I(a_N)\} = 5.28 \cdot 10^{-14}. \tag{5.22}$$

These are heuristic extrapolations far below the resolution of the uniform sampler. The straight line through the two cutoffs having positive hit counts has slope -2.38467; with only two points it carries no residual regression degrees of freedom. A descriptive six-cutoff binomial likelihood fit gives exponent 2.45306 and profile interval $[2.04052, 2.94753]$.

6. Classical analytic meaning of the quadratic scale

6.1. **Full-prefix and truncated normalizations.** The classical full-prefix quantity is

$$\mathfrak{Q}(N) = \frac{1}{\log N} \sum_{n \le N} \frac{M(n)^2}{n^2}. \tag{6.1}$$

For fixed N_0, set

$$K_{N_0} = \sum_{n \le N_0} \frac{M(n)^2}{n^2}. \tag{6.2}$$

Then the exact conversion is

$$\mathfrak{Q}(N) = \frac{K_{N_0} + L_N \mathcal{Q}_{N_0,N}(\mu)}{\log N}. \tag{6.3}$$

The two normalizations have the same limit whenever either limit exists, but they can be far apart at finite N. With $N_0 = 1000$,

$$K_{1000} = 2.068991, \qquad \mathfrak{Q}(455052511) = 0.122952, \tag{6.4}$$

whereas the truncated value is 0.029334. Thus the observed flatness of the latter is a property of the fixed lower cutoff, not evidence that the full-prefix normalization has numerically converged.

6.2. **Weak Mertens and RH.** For integer N,

$$\int_1^{N+1} \left(\frac{M(x)}{x}\right)^2 dx = \sum_{n=1}^{N} \frac{M(n)^2}{n(n+1)} = \sum_{n=1}^{N} \frac{n}{n+1}\frac{M(n)^2}{n^2}, \qquad \frac{1}{2} \le \frac{n}{n+1} < 1. \tag{6.5}$$

Hence boundedness of $\mathfrak{Q}(N)$ is equivalent, up to absolute comparison constants, to the weak Mertens conjecture

$$\int_1^{X} \left(\frac{M(x)}{x}\right)^2 dx \ll \log X. \tag{6.6}$$

This conjecture implies RH and simplicity of all nontrivial zeros; see [24, 20, 19]. RH alone is not known to imply (6.6).

There is also a direct positive reformulation. Define

$$D(N) = \sum_{k \le N} \frac{\mu(k)^2}{k}, \tag{6.7}$$

$$C(N) = \sum_{k \le N} \frac{\mu(k) M(k-1)}{k}, \tag{6.8}$$

$$B(N) = D(N) + 2C(N). \tag{6.9}$$

Discrete summation by parts gives

$$B(N) = \frac{M(N)^2}{N} + \sum_{k<N} \frac{M(k)^2}{k(k+1)} \ge 0. \tag{6.10}$$

It follows directly from the classical Mertens criterion that

$$\text{RH} \quad \Longleftrightarrow \quad B(N) \ll_\varepsilon N^\varepsilon \quad \text{for every } \varepsilon > 0. \tag{6.11}$$

This is a reformulation, not a new criterion for RH.

Proposition 6.1 (Conditional meaning of a bounded class ratio)**.** *Assume*

$$\frac{M(N)^2}{Q(N)\log(N/N_0)} \longrightarrow 0, \tag{6.12}$$

equivalently $M(N) = o(\sqrt{N \log N})$*. Then*

$$\overline{\mathcal{Q}}_N \longrightarrow d = \frac{1}{\zeta(2)} \tag{6.13}$$

and

$$\sup_N \rho_Q(N) < \infty \quad \Longleftrightarrow \quad \mathcal{Q}_{N_0,N}(\mu) = O(1) \quad \Longleftrightarrow \quad \mathfrak{Q}(N) = O(1). \tag{6.14}$$

Proof. The elementary estimate $Q(n) = dn + O(\sqrt{n})$ gives $\Sigma_1 = dL_N + O(1)$. In (3.11), the bridge correction is $O(L_N^{-1})$, while the endpoint contribution is $O(M(N)^2/[Q(N)L_N])$. Equation (6.12) proves (6.13). The first equivalence in (6.14) follows because the mean is then bounded above and away from zero; the second follows from (6.3). □

Warning 6.2 (The endpoint hypothesis is not an RH consequence). The standard consequence of RH, $M(N) = O_\varepsilon(N^{1/2+\varepsilon})$, does not imply $M(N) = o(\sqrt{N \log N})$. Assuming the stronger $M(N) = O(\sqrt{N})$ would make boundedness of the quadratic mass immediate. Thus Proposition 6.1 cannot be invoked from RH alone.

Remark 6.3 (The status of an asymptotic five-percent claim). A statement such as

$$\rho_Q(N) \le 0.05 + o(1) \tag{6.15}$$

would be a quantitative strengthening of bounded logarithmic second moment, not an elementary consequence of the finite model. Even the weaker assertion $\sup_N \rho_Q(N) < \infty$ requires, under (6.12), the weak Mertens conjecture. The finite data do not prove (6.15); they only show ratios near five percent over the stated cutoffs.

6.3. **Conditional spectral constant.** Under RH together with the negative-moment hypothesis $J_{-1}(T) \ll T$, Ng proves the logarithmic second-moment limit [19]

$$\frac{1}{Y}\int_0^Y e^{-y} M(e^y)^2\, dy \longrightarrow \beta_M, \tag{6.16}$$

$$\beta_M = 2\sum_{\gamma>0} \frac{1}{|\rho\zeta'(\rho)|^2}. \tag{6.17}$$

A separate reported computation gives

$$2\sum_{k=1}^{400} \frac{1}{|\rho_k\zeta'(\rho_k)|^2} = 0.0287674, \tag{6.18}$$

$$\widehat{R}_{400} = \frac{2\widehat{c}}{T_{400}} = 0.0002715, \qquad \widehat{c} = 0.0923, \quad T_{400} = 679.74, \tag{6.19}$$

$$\widehat{\beta}_M = 0.0290389. \tag{6.20}$$

The second term is a tail ansatz, not a computed sum. It estimates the positive-ordinate density of $|\zeta'(\rho)|^{-2}$ from the last quarter of the 400 ordinates.

Gonek's conjecture is

$$J_{-1}(T) := \sum_{0<\gamma\le T} \frac{1}{|\zeta'(\rho)|^2} \sim \frac{3}{\pi^3} T, \tag{6.21}$$

where $3/\pi^3$, not $6/\pi^3$, is the constant for positive ordinates [9, 19]. Substitution into the same tail ansatz gives

$$\frac{2(3/\pi^3)}{T_{400}} = 0.0002847, \qquad 0.0287674 + \frac{2(3/\pi^3)}{T_{400}} = 0.0290521. \tag{6.22}$$

The empirical $\widehat{c} = 0.0923$ is 4.6 percent below $3/\pi^3 = 0.0967546\ldots$, so there is no factor-two discrepancy.

Under the hypotheses needed for (6.16) and for the null mean limit,

$$\rho_Q(N) \longrightarrow \rho_\infty = \zeta(2)\beta_M. \tag{6.23}$$

The tail-corrected 400-zero value gives

$$\widehat{\rho}_\infty = 0.0477671, \qquad I(\widehat{\rho}_\infty) = 2.37284. \tag{6.24}$$

This explains why a ratio near five percent is plausible. It also answers the novelty question sharply: the approximate asymptotic percentage is a direct conditional combination of a classical spectral constant with the fixed-class mean d; the finite exact denominator and its comparison with the observed word are the model-specific contribution.

The reported spectral mass is concentrated at low ordinates:

$$\text{first zero: } 54.7\%, \qquad \text{first five: } 80.3\%, \qquad \text{first one hundred: } 97.4\%. \tag{6.25}$$

Thus the small quadratic scale is principally a low-frequency property. The same separate computation reports a zero-parameter reconstruction of the fourteen truncated values with RMS error $1.5\cdot 10^{-6}$. These spectral numbers require the separate reproducibility package specified in Appendix B; they are not inputs to Proposition 5.1.

Warning 6.4 (Quadratic mass does not control remote maxima). The constant in (6.17) is an ℓ^2-type norm of zero coefficients. It does not bound their ℓ^1-norm, force phase alignment, or control $\sup_x |M(x)|/\sqrt{x}$. Finite flatness of $\mathcal{Q}_{1000,N}$ therefore gives no evidence that the normalized Mertens maximum remains globally bounded.

7. Structural experiments beyond quadratic mass

The tests in this section ask whether the observed order contains information not already summarized by the quadratic functional. Their status is experimental, and their numerical specifications are recorded in Appendices A and B.

7.1. **Excursion orientation and shape.** For each completed excursion, let ε_r be its orientation, ℓ_r its support-time length, and

$$s_r = \sum_{n \text{ in excursion } r} \frac{1}{n} \tag{7.1}$$

its logarithmic integer duration. The tested orientation statistics were

$$\Theta = \frac{\sum_r s_r \varepsilon_r}{\sum_r s_r}, \tag{7.2}$$

$$P_1 = \frac{1}{K-1} \sum_{r<K} \varepsilon_r \varepsilon_{r+1}, \tag{7.3}$$

$$A_\ell = \mathrm{Corr}(\varepsilon_r, \log \ell_r). \tag{7.4}$$

Conditional randomization with the unsigned excursion skeleton fixed gave

$$p_\Theta = 0.574, \qquad p_{P_1} = 0.271, \qquad p_{A_\ell} = 0.479. \tag{7.5}$$

No orientation bias, nearest-neighbour persistence, or sign–length association was detected.

For unsigned excursion shapes, the tested coordinates were

$$A_r^{\mathrm{sh}} = \frac{1}{\ell_r^2} \sum_t H_r(t)^2, \tag{7.6}$$

$$H_r^{\max} = \frac{\max_t H_r(t)}{\sqrt{\ell_r}}, \tag{7.7}$$

$$\mathrm{HS}_r = \text{Horton–Strahler order}. \tag{7.8}$$

Five hundred observed excursions and 100 conditioned Catalan replicates gave

$$p_A = 0.505, \qquad p_H = 0.743, \qquad p_{\mathrm{HS}} = 0.485. \tag{7.9}$$

The tested shape statistics therefore do not distinguish the observed sample from the conditioned Catalan reference. Catalan enumeration and uniform Dyck paths are standard; see [23, 8].

7.2. **Scale partition and slow-envelope controls.** The partition audit in Table 6 compares an unweighted bin standard deviation with a Pearson-normalized statistic. The former has a two-sided value $p = 0.0066$ at 24 bins but not at 12 or 48 bins; the latter is nonsignificant at all three resolutions. There is therefore no partition-robust result.

For sliding windows, the short-scale statistics at window sizes 256 and 512 have correlations

$$r_{256} = 0.931, \qquad r_{512} = 0.914, \qquad p < 10^{-4} \quad \text{in both cases} \tag{7.10}$$

with the local trend $M(x)/\sqrt{x}$. At window size 1024, the reported split-half variance ratios are

$$(0.248,\ p = 0.111) \quad \text{and} \quad (0.706,\ p = 0.687). \tag{7.11}$$

The familywise result over seven scales from 256 through 2978 is

$$p_{\mathrm{global}} = 0.0645. \tag{7.12}$$

These controls identify the smaller-scale feature with the known slow Mertens envelope and show that the larger-scale feature does not replicate between halves.

7.3. **Conditional-choice analysis.** For blocks of q nonzero entries, the feature set included second- and third-order Rényi entropies, zero local time, self-intersection count, path width, turn rate, the quadratic functional, transposition stability, and their prescribed interactions. Given a choice set $\mathcal{S}_r$, the conditional-logit model was

$$\mathbb{P}(\pi_j \mid \mathcal{S}_r) = \frac{\exp(\beta^{\mathsf{T}} X_{rj})}{\sum_{\ell} \exp(\beta^{\mathsf{T}} X_{r\ell})}. \tag{7.13}$$

See [17] for conditional logit and [2] for false-discovery control.

At $q = 256, 512, 1024$, the quadratic-only deviance tests gave

$$p_{\mathrm{ANOVA}} = (0.550,\ 0.904,\ 0.661), \tag{7.14}$$

with held-out gains per set

$$(1.0 \cdot 10^{-6},\ -4.16 \cdot 10^{-4},\ 0.00245). \tag{7.15}$$

The complete feature hierarchy did not produce a robust held-out selector. These results are distinct from the blind spectral experiment below.

7.4. **Blind recovery of logarithmic frequencies.** Define the log-time signal

$$Y_\theta(t) = e^{-\theta t} M(\lfloor e^t \rfloor). \tag{7.16}$$

A hierarchical spectral procedure chose θ and frequencies from the Mertens path without reading zeta zeros or their derivatives. The external reference audit was performed only after fitting. Before the permutation null filter, the procedure matched 10 of the first 14 reference ordinates. For the first four it reported

reference ordinate	14.1347	21.0220	25.0109	30.4249
blind estimate	14.138	21.017	25.011	30.430

. (7.17)

The full fourteen-mode audit appears in Table 8.

The complete discovery, validation, and test pipeline was then rerun on 100 uniform fixed-mask permutations. A mode-by-mode filter at $q = 0.010$ retained

$$0 \text{ of the } 11 \text{ pre-null candidate modes.} \tag{7.18}$$

With the add-one empirical convention, however, the smallest attainable modewise value and the first Benjamini–Hochberg threshold were

$$\widehat{p}_{\min} = \frac{1}{100+1} = 0.00990099\ldots, \qquad \frac{q}{11} = 0.00090909\ldots. \tag{7.19}$$

Thus the design had no resolution for a sparse rank-one discovery at the declared q. At least

$$R \geq \left\lceil \frac{11}{0.010} \right\rceil - 1 = 1099 \tag{7.20}$$

complete placebo reruns are required merely to make the first BH threshold attainable. A prespecified global maximum statistic would answer a different familywise question and could use the 100 reruns more efficiently.

Remark 7.1 (Interpretation of the blind spectrum). The pre-null frequency recovery is real and should not be omitted, but it is not a discovery of previously unknown zeros. Classical explicit formulae place zero ordinates in the log-frequency content of $M(x)/\sqrt{x}$; a successful blind fit is therefore an algorithmic recovery of known spectral structure. The observation that no candidate survived the 100-placebo modewise filter is inconclusive, because (7.19) makes that filter underpowered by construction for a sparse discovery. The run neither establishes a new fixed-class spectral effect nor supplies strong evidence that no such effect exists. It is also not the ANOVA in (7.14): the two procedures answer different questions.

8. A route toward genuine arithmetic selection

8.1. **The exact selector already known.** The fixed-mask class discards the arithmetic relation that uniquely defines the Möbius function. That relation can be restored explicitly. For a word w with $w(1) = 1$, define the divisor-convolution defect

$$\Delta_N(w) = \sum_{n=2}^{N} \omega_n \left| \sum_{d|n} w(d) \right|^2, \qquad \omega_n > 0. \tag{8.1}$$

Proposition 8.1 (Unique finite arithmetic selector)**.** *For every $N \geq 2$,*

$$\Delta_N(w) = 0 \quad \Longleftrightarrow \quad w(n) = \mu(n) \text{ for all } n \leq N. \tag{8.2}$$

Proof. The left side is equivalent to $\sum_{d|n} w(d) = 0$ for every $2 \leq n \leq N$, together with $w(1) = 1$. These are the finite equations $(w * \mathbf{1})(n) = \delta_{n1}$. The Dirichlet inverse of the constant-one function is unique and equals μ. Equivalently, induction uses

$$w(n) = -\sum_{\substack{d|n \\ d<n}} w(d) \tag{8.3}$$

to determine each value. □

This proposition gives a unique position to M in the strongest possible sense, but it merely restates Möbius inversion. It contains no cancellation estimate and therefore supplies no progress toward RH.

8.2. **Why the finite class cannot classify RH behavior.** RH is a statement about one projectively consistent infinite arithmetic sequence. A member of $\Omega_N^{(1)}$ is only a finite word, and the uniform classes for successive N do not define a canonical infinite extension. It is therefore not mathematically meaningful to partition one finite class into words that satisfy or violate RH without first prescribing an extension law.

For an infinite sequence w, one may define its growth exponent

$$h(w) = \limsup_{x\to\infty} \frac{\log(1 + |R_w(x)|)}{\log x}. \tag{8.4}$$

For the Möbius function, RH is equivalent to $h(\mu) \leq 1/2$. The stronger weak-Mertens property is

$$\sup_{N\geq 2} \frac{1}{\log N} \sum_{n\leq N} \frac{R_w(n)^2}{n^2} < \infty. \tag{8.5}$$

For arbitrary paths, (8.5) does not control the supremum because sparse spikes can be hidden in an average. Its implication for RH is specifically arithmetic and uses the Mellin transform of M and $1/\zeta$.

8.3. **The missing inequality.** A nontrivial selection theorem would need more than $\Delta_N(\mu) = 0$. One viable target is a scale-consistent inequality that links a relaxed divisor defect to a positive path functional, schematically

$$B_w(N) = \frac{R_w(N)^2}{N} + \sum_{k<N} \frac{R_w(k)^2}{k(k+1)} \tag{8.6}$$

and

$$B_w(N) \leq C_\varepsilon N^\varepsilon + \mathcal{R}_N(\Delta_{\leq N}(w)), \qquad \mathcal{R}_N(0) = 0, \tag{8.7}$$

where B_w is the positive quantity analogous to (6.10). Applied to $w = \mu$, such a bound would imply the RH criterion (6.11). No inequality of the required strength is proved here; obtaining it would be at least as difficult as the target analytic estimate.

A statistically meaningful intermediate programme has three parts:

(i) condition simultaneously on the mask, endpoint, and quadratic-mass profile over many nested cutoffs, rather than on one scalar value;
(ii) test coordinates orthogonal to the absolute path, such as orientation bits and cross-level nesting, under exact conditional nulls;
(iii) require projective consistency in N and a deterministic link from any surviving coordinate to the divisor-convolution defect or to (6.10).

The orientation and shape tests in Section 7 do not detect a departure on the available interval. The blind spectrum recovers known analytic structure but does not provide an independent combinatorial selector. Thus no current statistic uniquely distinguishes the Möbius path from all RH-compatible paths, let alone from paths satisfying a stronger mean-square property.

Question 8.2 (Projectively consistent residual selector)**.** Does there exist a statistic $\mathcal{S}_N(w)$, not determined by the mask, endpoint, or multiscale quadratic-mass profile, whose conditional null is explicit, whose value is consistent under increasing cutoffs, and for which a deterministic inequality links $\mathcal{S}_N(\mu)$ to (6.11)?

9. Conclusion

The model has a rigorous finite core and a limited analytic interpretation. The fixed-mask, fixed-count orbit is exactly specified; its support-time path is a hypergeometric bridge at the square-root critical scale. The normalized logarithmic quadratic path mass has exact finite mean and variance, an exact order-dependent bilinear term, and an exact transposition calculus. On the identified data, the Möbius word has 5.18 percent of its class mean at the largest cutoff, with the inequality separated from zero by a large forward-error margin.

For an expert in analytic number theory, the five-percent magnitude is not a new asymptotic law: under Ng's hypotheses it is explained by $\zeta(2)\beta_M$, and the first few zeros dominate β_M. What the model adds is a finite conditional baseline and a set of exact combinatorial coordinates for asking how the sign order differs from that baseline. The reported lower tail is much thinner than a two-moment approximation, but its exact probability remains unknown.

No tested excursion orientation, Catalan shape, scale partition, or local conditional-choice statistic supplies a surviving selector beyond the quadratic scale. The blind spectral procedure recovers several known zero frequencies without using them in the fit, but its permutation-corrected mode filter retains none; with only 100 placebos, that filter is too coarse for a sparse BH discovery and the outcome is therefore inconclusive. The only exact unique selector presently available is the divisor-convolution identity, and connecting that identity to square-root cancellation is the unresolved arithmetic problem rather than a consequence of the finite permutation model.

Appendix A. Numerical tables

This appendix collects the numerical outputs used in the text. Exact finite-class formulae are distinguished from sampled statistics in every caption.

A.1. **Fixed-class quadratic comparison.** The complete-cutoff support contains 276,638,778 nonzero entries. The corresponding squarefree density is 0.60793 to the displayed precision. All fourteen parity checks

$$M(N) \equiv Q(N) \pmod 2 \tag{A.1}$$

passed.

TABLE 1. Observed quadratic path mass and exact fixed-mask, fixed-count class moments, with $N_0 = 1000$. Decimal entries are binary64 evaluations of the exact formulae.

N	$M(N)$	$\mathcal{Q}(\mu)$	$\overline{\mathcal{Q}}_N$	σ_N	ρ_Q	Z_Q
10,000	−23	0.032174	0.39129	0.37891	0.082224	−0.9478
30,000	18	0.030803	0.43847	0.39324	0.070251	−1.0367
100,000	−48	0.031084	0.48239	0.39632	0.064437	−1.1387
300,000	220	0.029369	0.53006	0.39912	0.055407	−1.2545
1,000,000	212	0.030100	0.52665	0.37332	0.057154	−1.3301
5,000,000	−709	0.029515	0.54848	0.35510	0.053812	−1.4614
50,000,000	−908	0.029733	0.55336	0.32471	0.053732	−1.6126
100,000,000	1,928	0.029641	0.55844	0.31817	0.053079	−1.6620
150,000,000	1,089	0.029536	0.55767	0.31333	0.052963	−1.6855
200,000,000	320	0.029772	0.55824	0.31038	0.053332	−1.7027
250,000,000	−982	0.029944	0.55941	0.30838	0.053529	−1.7169
300,000,000	−592	0.029662	0.55989	0.30663	0.052977	−1.7292
350,000,000	2,896	0.029508	0.56226	0.30578	0.052482	−1.7423
455,052,511	−5,414	0.029334	0.56628	0.30436	0.051800	−1.7642

TABLE 2. Uniform fixed-count sampling. Positive-hit intervals are Wilson intervals [25]; zero-hit entries are exact one-sided 95-percent Clopper–Pearson bounds.

N	draws	hits	$\widehat{P}_Q$	95-percent interval or bound	Z_s
10,000	300,000	412	0.0013733	[0.0012470,0.0015124]	1.3216
30,000	150,000	15	0.0001000	[0.0000606,0.0001650]	0.9930
100,000	100,000	0	0	$< 2.9957 \cdot 10^{-5}$	−1.6256
300,000	50,000	0	0	$< 5.9913 \cdot 10^{-5}$	−0.7217
1,000,000	10,000	0	0	$< 2.9953 \cdot 10^{-4}$	−1.9583
5,000,000	10,000	0	0	$< 2.9953 \cdot 10^{-4}$	−1.2327

A.2. **Uniform-permutation ladder.** Here $Z_s = (s - \sigma_N)/\operatorname{SE}(s)$, with $\operatorname{SE}(s)$ estimated from the sample fourth moment. The first 8,200 draws at $N = 5 \cdot 10^6$ have

$$\overline{\mathcal{Q}}_{\text{sample}} = 0.544194764161, \qquad s_{\text{sample}} = 0.350499677391. \tag{A.2}$$

These are compatible with the exact mean 0.548481916755 and exact standard deviation 0.35510. The value 0.350195 is the leading $L_N - 5/2$ asymptotic standard deviation, not the exact finite-class value.

A.3. **Support-time coefficients.** The chosen reference

$$c_0 = \frac{0.0300}{6/\pi^2} = \frac{\pi^2}{200} = 0.049348022005\ldots \tag{A.3}$$

was fixed by the rounded target 0.0300, not estimated independently. The first fitted log-scale mode was

$$g_1(t) = -0.000774584 - 0.009187853\cos(2.541179t) - 0.006146902\sin(2.541179t), \qquad t = \log_2 K. \tag{A.4}$$

Its period is 2.47255 octaves, amplitude 0.011054, and in-sample $R^2 = 0.643529$. The complete causal forecast did not consistently improve on a constant predictor.

TABLE 3. Recorded scale coefficients. The second and third comparison columns arise from the feedback model and its scale adjustment; they are not independent estimates of $\zeta(2)\mathcal{Q}$.

K	c_{observed}	$c_{a=4.8161}$	$c_{a,\text{adjusted}}$
256	0.048729	0.047858	0.064982
512	0.067637	0.054504	0.066281
1024	0.036207	0.048046	0.065380
2048	0.049998	0.053648	0.037300
4096	0.054833	0.043456	0.050333
8192	0.044089	0.045434	0.046309
16,384	0.049748	0.050261	0.040051
32,768	0.038838	0.043234	0.044990
65,536	0.061641	0.050747	0.034831
131,072	0.039820	0.047357	0.062011
262,144	0.036670	0.047253	0.036656
524,288	0.058102	0.046882	0.035871

The separate spectral coefficient estimates are

K	direct	cross-fitted
14	0.11802	0.11793
50	0.10508	0.10508
100	0.10250	0.10270
102	0.10211	0.10231

. (A.5)

No equality between these spectral coefficients and (3.33) is asserted.

A.4. **Excursion diagnostics.** At $N = 5 \cdot 10^6$,

$$m = 3{,}039{,}633, \qquad p = 1{,}519{,}462, \qquad q = 1{,}520{,}171. \tag{A.6}$$

There are 6,016 completed excursions in the full support-time path, so

$$\log_{10} |\mathcal{O}_N(\mu)| = 6016 \log_{10} 2 = 1810.996\ldots. \tag{A.7}$$

A windowed analysis excluding excursions meeting $n \le 1000$ contains 5,956 completed excursions.

A reported sample of 300 uniform bridges gave

$$\overline{Z} = 1679.8, \qquad s_Z = 1102.4, \qquad \min Z = 7, \qquad \max Z = 5544, \tag{A.8}$$

so none reached the observed value 6,016. The corresponding exact one-sided 95-percent zero-exceedance bound is

$$1 - 0.05^{1/300} = 0.00994. \tag{A.9}$$

For the longest completed-excursion fraction, the same sample had mean 0.5301, median 0.4806, and minimum 0.1219, compared with the observed 0.11737.

TABLE 4. Observed excursion-length tails and the unconditioned first-return benchmark $6016\sqrt{2/\pi}\, c^{-1/2}$.

threshold c	observed	benchmark	ratio
10	1474	1517.9	0.971
100	454	480.0	0.946
1000	140	151.8	0.922
10,000	40	48.0	0.833

TABLE 5. Large-cutoff support and return census. The complete-cutoff return count was not recorded in the cited census.

N	nonzero entries	returns to zero	$M(N)$
50,000,000	30,396,344	15,756	−908
100,000,000	60,792,694	25,525	1,928
150,000,000	91,189,087	34,232	1,089
200,000,000	121,585,426	36,069	320
250,000,000	151,981,736	36,755	−982
300,000,000	182,378,126	42,505	−592
350,000,000	212,774,472	47,215	2,896
455,052,511	276,638,778	not recorded	$-5,414$

Large computations of the Mertens function provide numerical context for these finite ranges [5, 16, 15].

TABLE 6. Partition audit. Each p-value is two-sided; the Pearson statistic has approximately one fewer degree of freedom than the number of bins.

bins	unweighted: observed/null	p	Pearson: observed/null	p
12	0.252/0.352	0.135	6.07/11.06	0.188
24	0.351/0.512	0.0066	15.65/23.24	0.103
48	0.706/0.710	0.837	45.89/47.56	0.739

TABLE 7. Held-out audit of the local quadratic-only conditional-choice model. Each scale used 1,000 choice sets with 13 paths.

q	deviance p	test gain per set	test top–1
256	0.550	$1.0 \cdot 10^{-6}$	0.120
512	0.904	$-4.16 \cdot 10^{-4}$	0.040
1024	0.661	0.00245	0.120

A.5. **Partition and conditional-choice diagnostics.**

A.6. **Blind frequency audit.**

APPENDIX B. REPRODUCIBILITY SPECIFICATION

This appendix specifies the complete deposit required to reproduce every computational statement in the paper. It is a protocol, not a claim that an independent rerun was performed while preparing this version. The exact finite identities can be checked without the data; the decimal tables require the identified inputs and the declared software environment.

B.1. **Required archive structure.** The reproducibility archive shall contain the following top-level items:

(i) `README.md`, with the single-command instructions, expected wall time, memory requirement, and a map from each table and equation in the paper to its raw output;
(ii) `MANIFEST.sha256`, containing a SHA-256 digest for every input, source file, raw output, and generated table;

TABLE 8. External audit of the first fourteen zero ordinates against the pre-null blind spectral candidates. Reference values were not used during fitting. A dash denotes no matched candidate.

mode	reference	blind estimate	error
1	14.135	14.138	0.00347
2	21.022	21.017	−0.00461
3	25.011	25.011	0.00048
4	30.425	30.430	0.00471
5	32.935	32.931	−0.00422
6	37.586	37.559	−0.02737
7	40.919	–	–
8	43.327	–	–
9	48.005	48.015	0.00970
10	49.774	49.837	0.06298
11	52.970	–	–
12	56.446	–	–
13	59.347	59.377	0.03029
14	60.832	60.810	−0.02195

(iii) `environment.txt`, recording operating system, processor, available memory, MATLAB release and update, BLAS library, thread count, and locale;
(iv) `data/moebius.bin`, the signed-byte Möbius word;
(v) `data/zeta_zeros_400.csv` and `data/zeta_derivatives_400.csv`, used only by the separate spectral branch;
(vi) `src/`, containing the canonical source files listed below;
(vii) `raw/`, containing unmodified console logs, CSV files, and MAT files from every run;
(viii) `tables/`, containing only deterministic transformations of the raw files into manuscript tables;
(ix) `tests/`, containing the finite exhaustive tests and their expected outputs.

No source file may download data during a run. The exact/permutation branch and the spectral branch must have separate entry points, and the former must abort if a spectral input is opened.

B.2. **Canonical source inventory.** The archive shall use mathematical names and contain, at minimum,

`run_all_reproductions.m`:
Top-level orchestrator; runs tests, exact calculations, sampling experiments, and table generation in a fixed order.

`fixed_mask_moments.m`:
Streaming observed sum, exact class mean, exact variance, forward-error audit, continuum diagnostics, and all fourteen cutoffs.

`uniform_bridge_ladder.m`:
Uniform fixed-count sampler at the six Monte Carlo cutoffs.

`excursion_statistics.m`:
Completed-excursion decomposition, reflection count, exact return mean, orientation randomization, Catalan controls, and length diagnostics.

`partition_and_scale_tests.m`:
Audits with 12, 24, and 48 bins. Sliding scales, slow-envelope correlations, split halves, and familywise calibration.

`conditional_choice_analysis.m`:
Feature construction, conditional-logit models, chronological splits, and deviance tables.

`blind_log_spectrum.m`:
Blind discovery/validation/test spectral pipeline and the 100 complete fixed-mask placebo reruns.

`spectral_second_moment.m`:
Four-hundred-zero direct sum, tail alternatives, fourteen-cutoff reconstruction, and residual decomposition.

`make_manuscript_tables.m`:
Reads raw outputs and writes tables without recomputing any statistic.

Every function used by these entry points must be included as source rather than supplied through an unrecorded local path. The archive must also state the license under which the sources and derived tables are released.

B.3. **Input definition and integrity checks.** The main data file has the immutable specification

$$\begin{aligned} \text{format} &= \texttt{int8}, \\ \text{allowed values} &= \{-1, 0, 1\}, \\ \text{number of bytes} &= 455{,}052{,}511, \\ \text{SHA-256} &= \texttt{0e551f47d924ef6f3e348e131fdf53e0c677ad5e5471ef9d6249b3fd662e2ea4}. \end{aligned} \tag{B.1}$$

Before any statistic is evaluated, the program must:

(a) verify the byte count, allowed alphabet, and SHA-256 digest;
(b) independently sieve $\mu(n)$ for $1 \le n \le 10^6$ and compare every entry;
(c) verify at each cutoff the nonzero count and endpoint recorded in the raw reference table;
(d) abort unless
$$M(N) \equiv Q(N) \pmod 2 \tag{B.2}$$
at every cutoff;
(e) verify $p = (m+S)/2$, $q = (m-S)/2$, and $T_m = S$.

The spectral CSV files must contain one row per positive ordinate with columns `index`, `gamma`, `ReZetaPrime`, `ImZetaPrime`, `AbsZetaPrime`, working precision, software used, and an independent residual or interval check for $\zeta(1/2 + i\gamma)$. Their hashes must be listed in `MANIFEST.sha256`. The code must recompute $|\zeta'(\rho)|$ from its real and imaginary parts and abort if it disagrees with the stored absolute value beyond the declared tolerance.

B.4. **Exact-layer configuration.** The main run uses

$$N_0 = 1000, \tag{B.3}$$

$$\begin{aligned} \mathcal{N}_{\text{exact}} = \{&10^4, 3\cdot 10^4, 10^5, 3\cdot 10^5, 10^6, 5\cdot 10^6, \\ &5\cdot 10^7, 10^8, 1.5\cdot 10^8, 2\cdot 10^8, 2.5\cdot 10^8, 3\cdot 10^8, 3.5\cdot 10^8, 455052511\}. \end{aligned} \tag{B.4}$$

The streaming read block is 2^{22} bytes and the positive-sum audit block is 2^{16} terms. Products that can exceed 2^{53} must not be formed. In particular, the implementation shall use

$$\left(\frac{M(n)}{n}\right)^2, \qquad \frac{Q(n)}{n}\frac{Q(n)-1}{n}, \qquad \frac{Q(n)}{n}\frac{1}{n} \tag{B.5}$$

rather than forming $M(n)^2$, $Q(n)^2$, or n^2 first at the largest cutoffs.

The mean must be computed independently in two ways:

(a) from the streaming sums Σ_1, Σ_2 in (3.7)–(3.8);
(b) from the explicit plateau weights and (3.11).

The variance must likewise be evaluated by the prefix formula (3.13) and, at $N = 10^4$, by the literal triangular $O(m^2)$ sum (3.14).

B.5. **Floating-point audit.** The recorded main environment is MATLAB 24.2.0.2740171 (R2024b Update 1). The forward-error argument assumes IEEE-754 binary64, round-to-nearest ties-to-even, no extended precision, no fused-operation contraction, and no reassociation. If the execution environment does not satisfy these conditions, the program must label the audit inapplicable or replace it with directed-rounding interval arithmetic.

Within each block of r positive summands, ordinary summation is bounded using

$$\gamma_r = \frac{ru}{1-ru}, \qquad u = 2^{-53}, \tag{B.6}$$

and block totals are accumulated with compensated summation. Bounds are propagated through the algebra for $\overline{\mathcal{Q}}_N$ and $\Delta_N = \mathcal{Q}(\mu) - \overline{\mathcal{Q}}_N$. The run must abort unless

$$0 \notin [\Delta_N^-, \Delta_N^+] \quad \text{and} \quad \Delta_N^+ < 0 \tag{B.7}$$

at every cutoff. This procedure is an a-priori forward-error audit in the sense of [11]; it is not interval arithmetic with directed rounding.

B.6. **Mandatory exact self-tests.** The test suite must run before the data file is processed and must abort on any failure. It includes:

(i) complete enumeration of every fixed-count sign class with $1 \le m \le 8$, covering 56 classes and 1,890 covariance pairs;
(ii) comparison of every enumerated value $\mathrm{Cov}(T_j^2, T_k^2)$ with (2.15); the recorded maximum absolute error is $3.2 \cdot 10^{-13}$;
(iii) explicit enumeration of the 20 words with $m = 6, p = q = 3$, including $\mu_2 = -1/5$, $\mu_4 = 1/5$, and

$$\mathrm{Var}(T_2^2) = 3.84; \tag{B.8}$$

(iv) checks that $\psi(m) = 0$ and $\mathbb{E}T_m^2 = S^2$ identically;
(v) equality of the $O(m)$ and $O(m^2)$ variance implementations at $N = 10^4$, with recorded relative discrepancy $3.05 \cdot 10^{-16}$;
(vi) equality of the streaming and plateau-weight mean and variance routes, with recorded mismatches between $7 \cdot 10^{-18}$ and 10^{-16};
(vii) exhaustive checks of Proposition 4.3 for at least $(p, q) = (3, 3), (4, 3), (5, 4), (6, 4)$;
(viii) a SHA-256 unit test against a published standard test vector;
(ix) a symbolic or high-precision numerical self-test of the Legendre transform in (5.19), explicitly labelled as a transform test rather than a proof of the OU transfer.

The reference arithmetic printed by the test suite is

$$1 - 0.05^{1/2800} = 0.00106933223978, \tag{B.9}$$

$$(1 - 0.00428)^{2800} = 6.08478495942 \cdot 10^{-6}, \tag{B.10}$$

$$1 - 0.05^{1/9500} = 0.000315290524814, \tag{B.11}$$

$$9500(0.00428) = 40.66. \tag{B.12}$$

B.7. **Uniform sampling protocol.** The Monte Carlo configuration is

N	10^4	$3 \cdot 10^4$	10^5	$3 \cdot 10^5$	10^6	$5 \cdot 10^6$
R	300000	150000	100000	50000	10000	10000
seed	19092021	19092022	19092023	19092024	19092025	19092026

. (B.13)

The generator is MT19937, invoked as `rng(seed,'twister')` separately at each cutoff. A draw is produced by a uniform random permutation of the p positive and q negative signs, followed by placement on the fixed support. The program must check $T_m = S$ on every draw and test the marginal positive-sign frequency at fixed support indices.

For a positive hit count, the archive reports the Wilson interval and the exact binomial likelihood. For zero hits, it reports the one-sided exact bound

$$p_{\mathrm{upper}} = 1 - \alpha^{1/R}, \qquad \alpha = 0.05. \tag{B.14}$$

The sample standard-deviation check uses

$$\mathrm{SE}(s) = \frac{s}{2}\sqrt{\frac{\widehat{\kappa} - 1}{R}}, \qquad \widehat{\kappa} = \frac{\widehat{m}_4}{s^4}. \tag{B.15}$$

The program must print $(s - \sigma_N)/\,\mathrm{SE}(s)$ and require its absolute value to be below the declared calibration threshold.

The two-moment comparison uses

$$\nu = \frac{2\overline{\mathcal{Q}}_N^2}{\sigma_N^2}, \qquad \mathcal{Q}_{\chi^2} = \frac{\overline{\mathcal{Q}}_N}{\nu}\chi_\nu^2. \tag{B.16}$$

It is a deliberately falsifiable approximation, not the sampling law. Every sampled quadratic value, not only summary statistics, must be written to a per-cutoff CSV file.

MT19937 has 19,937 state bits and cannot visit every member of the largest class. The archive shall state this limitation. The exact mean and variance are unaffected; the Monte Carlo results concern the implemented uniform finite-permutation algorithm and require an independent generator as a robustness check before publication.

B.8. **Excursion and conditional-choice protocols.** The excursion archive must include:

(a) the complete ordered list of return times and completed-excursion lengths;
(b) a flag showing whether the terminal segment is incomplete and excluded from reflection;
(c) 100,000 orientation-null realizations with the unsigned skeleton fixed;
(d) the 500 observed excursions and all 100 Catalan replicates used for the shape tests;
(e) all 300 uniform bridges used for the return-count and longest-excursion summaries;
(f) the exact seed and generator state for each component.

The conditional-choice archive must include all 1,000 choice sets with 13 paths at each $q = 256, 512, 1024$, the feature matrix before and after standardization, the chronological split indices, coefficient estimates, standard errors, log likelihoods, deviance increments, degrees of freedom, held-out log scores, and false-discovery adjustment. Expected cell counts must be reported for every chi-square diagnostic; where they are too small, an exact or permutation calibration is required.

B.9. **Blind spectral protocol.** The blind log-spectrum run uses $N = 5 \cdot 10^6$, 1,321 logarithmic samples, frequency ceiling 216.275, four hierarchical levels, and growing capacity

$$J(n) = \lfloor n^{0.38} \rfloor, \qquad J_{\max} = 15. \tag{B.17}$$

The chronological discovery, validation, and test endpoints are $0.45N, 0.65N, N$. Reference ordinates must not be loaded until the blind fit and all 100 placebo fits are complete. Each placebo must rerun the entire selection pipeline, including exponent choice, peak selection, frequency refinement, and validation. The archive must preserve:

(a) all 1,321 sampled log-time values and observed signals;
(b) every candidate peak before and after hierarchical selection;
(c) all 100 placebo candidate sets and test gains;
(d) the mode-by-mode multiple-testing table;
(e) the external reference audit generated only after selection;
(f) warnings, ranks, and condition numbers from every linear solve.

The pre-null match table is an algorithmic recovery result. The recorded 100-placebo run is a pilot resolution audit only: by (7.19), zero surviving modes cannot be used as confirmatory evidence of absence. A confirmatory modewise analysis at $q = 0.010$ must rerun the complete pipeline on at least 1,099 placebos, and should use more if stable tail estimation is required. Alternatively, the protocol may prespecify one global maximum statistic and calibrate that single statistic with the full-pipeline placebo distribution. The choice between these two designs must be made before a new run and recorded in the manifest.

B.10. **Separate spectral-second-moment protocol.** The spectral run must use arbitrary precision high enough that rounding in each term of (6.18) is negligible at the printed precision. It must output:

(a) all 400 individual values $2/|\rho_k \zeta'(\rho_k)|^2$ and their cumulative sums;
(b) the contributions of the first, first five, and first one hundred zeros;
(c) the precise last-quarter estimator defining $\widehat{c} = 0.0923$;
(d) both tail alternatives $\widehat{c}$ and $3/\pi^3$;
(e) the fourteen reconstructed quadratic-mass values, their residuals, and the zero-parameter RMS;
(f) the fit $R_N = A + B/L_N$, its covariance matrix, and the discrete sum–integral correction;
(g) all working-precision and zero-validation diagnostics.

No output from this branch may be merged with the no-zero exact run before both raw manifests have been written.

B.11. **Raw output inventory and acceptance criteria.** At minimum, the archive shall preserve

```
exact_layer.csv, permutation_ladder.csv, sampled_quadratic_values.csv,
continuum_diagnostics.csv, ou_quantiles.csv, excursions.csv,
partition_audit.csv, conditional_choice.csv, blind_modes.csv,
blind_placebos.csv, spectral_terms.csv.
```
(B.18)

The MAT files must retain every accumulator and configuration object needed to audit the CSV files. Console output must be captured verbatim.

A reproduction is accepted only if:

(i) every input and source hash matches the manifest;
(ii) all exhaustive and integrity tests pass;
(iii) the fourteen observed values, exact means, and exact variances reproduce within the stated forward-error bounds;
(iv) the sign of every separation interval is negative;
(v) raw Monte Carlo summaries reproduce from the declared seeds, and an independent generator gives compatible confidence intervals;
(vi) the numerical tables are generated only from raw files;
(vii) every table cell in the manuscript is traceable to one raw column and row;
(viii) a fresh LaTeX build completes without undefined references, missing citations, or overfull material that obscures data.

The intended single command is

```
matlab -batch "run_all_reproductions"
```
(B.19)

from the archive root. The recorded full exact and Monte Carlo run required 2,133.7 seconds in the stated environment. The command and all programs in this appendix are specified for an independent rerun; they were not executed as part of the editorial production of Version 3.

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